

CHAPTER 7

Apparel



0916CH07



Figure 7.1: Apparel with traditional embroidery is being adapted for high-fashion wear

Figure 7.1 shows women of a nomadic community in their colourful, traditional attire. Creating this apparel is extremely labour-intensive and the art of making it is passed down generations. Given the unique beauty of this traditional apparel, it has moved out of these communities to global fashion. Designers work with members of these communities to create apparel, like embroidered jackets, dresses and accessories.

In this chapter, you will learn about the following:

Technical drawing of the apparel	Selecting fabric based on properties	Selecting materials and tools
Making bill of materials	Practising stitches	Preparing fabric, marking and cutting
Stitching by hand and/or machine	Finishing, packaging and labelling	

7.1 Introduction

Food, shelter and clothing are the basic needs of human beings. From the earliest times, clothing has been used to protect the body from weather and the environment. The history of clothing began with the use of animal skins and natural materials for warmth and protection. In India, evidence of clothing can be traced back to the Sindhū–Sarasvatī civilisation.

Clothing today is much more than just a way to protect the body. What we wear reflects our personality and our culture.

Clothing also reflects the current fashion and trends of the time.

Given the importance of this work, the apparel industry has developed around the design and manufacturing of clothing. This industry has close linkages with the textile industry, which is involved in the production of fibre, yarn and fabric, along with the fashion industry, which specialises in creating unique, branded, occasion-specific clothing (Figure 7.2). To elaborate on these linkages, as synthetic fibres became popular, traditional weaving gave way to industrial looms but as traditional weaving again become fashionable, it was ‘revived’.



Figure 7.2: The textile, apparel and fashion industry are closely connected, and each influences the other

7.2 Process chart

7.2.1 Scoping work

Deciding the scope of the work means that decisions need to be taken regarding the following:

1. **Availability of resources:** List the materials and tools available, and experts who can guide you.
2. **What will be created:** Think about the apparel you want to create. The questions you need to ask are whether you have the required resources and whether you can complete the task in the given time.
3. **Understanding the need:** Some questions you need to ask are – Who will use the apparel? Will it be comfortable, strong and easy to maintain?

DID YOU KNOW?

Upcycling gives new life to old or unused clothes and fabrics by creatively redesigning them into useful and attractive products. An old pair of jeans can be transformed into a useful sling bag with decorative additions of patchwork and mirror work (Figures 7.3 and 7.4). Similarly, fabric scraps can be stitched together to make a colourful patchwork table runner or wall hanging.



Figure 7.3: *Turning denim jeans into a carry bag*



Figure 7.4: *Sewing pieces for patchwork*

7.2.2 Making a process chart

Once you have identified the apparel you will make, you can list all the tasks required to make the apparel, estimate the timelines for each task and who will take the responsibility for each of the tasks. Table 7.1 shows a sample process chart for making apparel.

Making a
process chart



Table 7.1: Process chart for making apparel

Tasks for making apparel	Dates	Responsibility
Selecting fabric based on properties		
Design and measurement		
Drafting and pattern making		
Fabric preparation and cutting		
Stitching apparel		
Finishing, packaging and labelling		

7.3 Site visit

Before beginning the work, it is important to observe and interact with experts (for example, tailor, fashion designer, boutique owner, garment worker, etc.) in the presence of a teacher. You can also take their guidance in planning the work undertaken by you.



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You can use the pointers in Table 7.2 for discussion with practitioners and take notes.

Table 7.2: Points for observation during site visit

Points of observation/ Discussions	Description
Tools and materials used	Materials used and their storage Tools used and their maintenance
Key processes	Key steps and their importance
Safety protocols	Using appropriate tools, safety precautions, etc.
Schedules	Frequency and timing of key tasks
Quality criteria	Criteria for quality inputs, process and output
Technology use	Digital tools/apps used

Think of any other points for observation while visiting the site. For example:

1. What do the practitioners value the most about their work (for example, quality of apparel, specific processes, pride in making something useful, etc.)?
2. You can ask about possible challenges you may face while making apparel and how to overcome them.

Once you have returned from the visit, work in groups to add details to the process chart for your work.

7.4 Technical drawing of apparel

The first step towards making the apparel is to visualise how your product will look. What will be the design? Will it have any accessories, such as buttons or zips? Or strings? What fabric will it be made of and which colour will be best suited? These are just examples of some questions you need to consider.

You also need to ensure the apparel fits – this will require measurement. Accurate measurement is important to make an apparel of the right size and shape. If you do not have a specific person in mind, you can ask an expert or learn about standard sizes through online sources. Apparel are generally made using these standard sizes when produced in bulk.

It is easy to measure something like a cushion cover – it is generally a square, rectangle or circle. But when it comes to apparel, measurements need to be taken differently (Figure 7.5).



Technical drawing

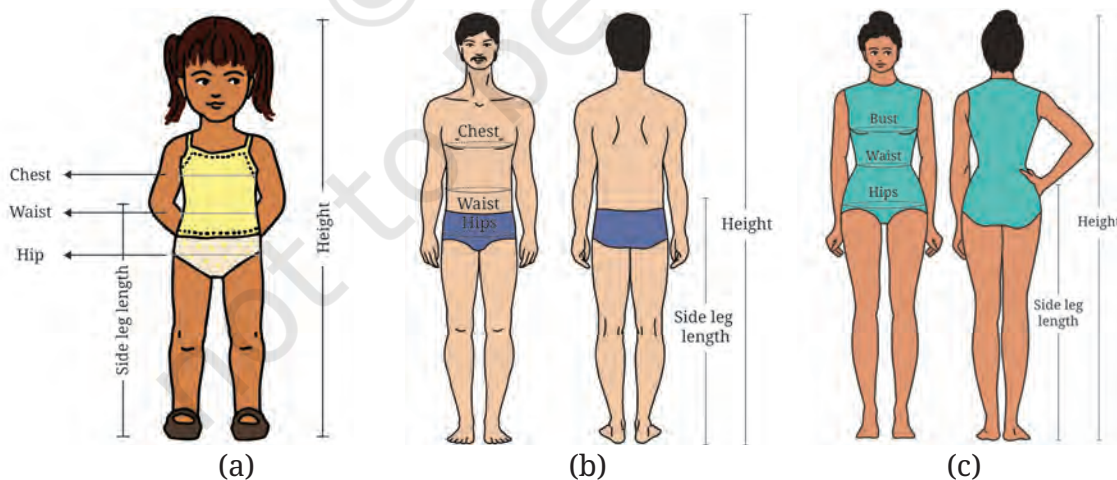


Figure 7.5: The images show the body points where measurement needs to be taken for stitching apparel for (a) a child, (b) an adult male and (c) an adult female



While apparel can be 'bespoke' or custom made for a specific customer, standard sizes are sold as ready-made apparel.



Quality

QUALITY

While taking measurements, the following need to be ensured:

1. The person is not wearing thick clothing.
2. Use a tailoring tape since it is flexible.
3. The person should stand straight and relaxed while you are taking the measurements.
4. Do not keep the tape either too loose or too tight.
5. Take chest/bust, waist, hips, shoulder, sleeve and length measurements as in Figure 7.5.

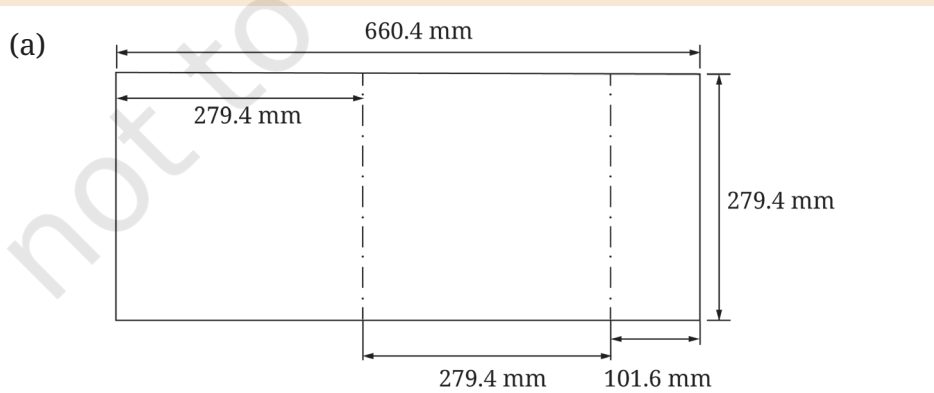
Remember, you will have to leave extra fabric for seams or finishing while marking the fabric – this is known as 'seam allowance'.



CASELET

Students of Grade 9 of Government High School divided themselves into three groups. Group 1 decided they would start with a furnishing product – they saw that the cushion covers in the waiting area outside the Principal's office had worn out and decided to replace them. Group 2 observed that the clothes of the person serving the midday meal were getting stained and decided to make an apron. Group 3 decided that they would make a dress for their younger siblings.

Three drawings – of a cushion cover, an apron and the dress of a young child – made by each of the three groups are given in Figure 7.6.



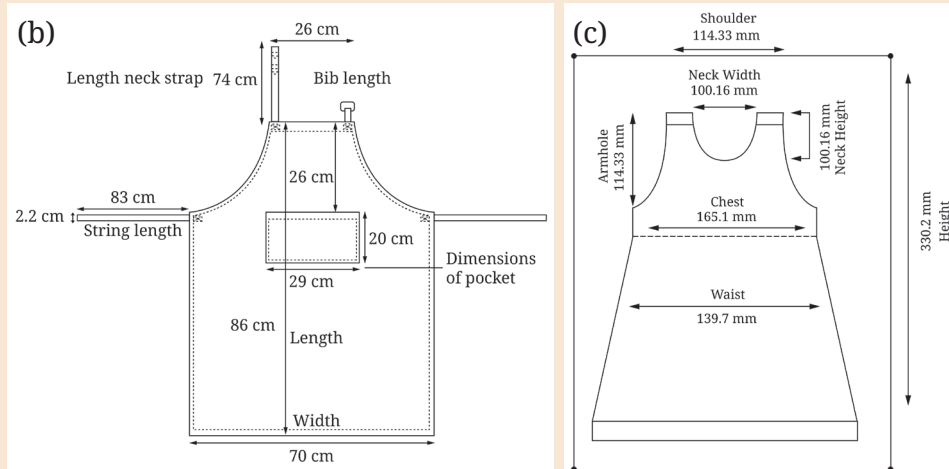
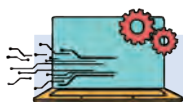


Figure 7.6: Sketch of (a) cushion cover, (b) apron and (c) dress for a young child



TECHNOLOGY AND ARTIFICIAL INTELLIGENCE

You can use computer-aided design (CAD) software to make the technical drawing of the selected apparel.



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Think of different design options for the apparel you will make. Take measurements carefully. Make a neat technical drawing, detailing out the exact measurements and label all parts clearly. Also, write notes about fabric, thread or special features like zippers, buttons or embroidery.

DID YOU KNOW?

Fabric for apparel is selected based on its characteristics (Figure 7.3).

Table 7.3: Examples of some fabrics and their characteristics

Fabric		Characteristics
Cotton		Soft, breathable and comfortable for daily wear. Suitable for everyday clothing.
Wool		Breathable and wrinkle free. Depending on the processing, some wool varieties are soft while others are rough, for example, sweaters, shawls, coats, trousers, <i>phirans</i> , etc.



Silk		Naturally soft and shiny. It is expensive compared to cotton, and usually used on auspicious and festive occasions. Some silk fabrics have temperature regulating properties.
Linen		One of the strongest natural fibres. It is ideal for summer clothing, home furnishing, etc. With proper care, the longevity can be ensured.



Select tools and materials

7.5 Selecting materials

You read in Chapter 5 that materials are selected for creating different kinds of products based on their properties. This is also true for apparel making. Table 7.4 lists some common fabric, their properties and the kind of apparel they are used for making.

You may select more than one kind of fabric depending on the apparel, for example, if you plan to use a different colour or print to make a pocket or for patchwork.

Besides the fabric, you will also need thread for stitching, preferably of the same colour as the fabric(s) you have chosen.

Table 7.4: Selection of fabric is done on the basis of the apparel they will be used to make

Apparel	Suitable fabric	Reason for selection
T-shirt	Cotton	Soft, breathable, comfortable for daily wear
School uniform	Polyester-cotton blend	Durable, easy to wash, wrinkle-resistant
Dress	Cotton, rayon	Soft, breathable, comfortable for daily wear
Baby dress	Soft cotton, muslin	Soft, breathable and gentle on the skin
Kitchen apron	Thick cotton, polyester	Strong, washable, protects clothes



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Which material did you select and why?

7.6 Selecting tools

The important tools required for making most apparel are given in Table 7.5 below.

Table 7.5: Commonly used tools for making apparel

Tools	Use	Safety note
Tailoring tape	To take measurements	Do not pull tightly around the body
Scissors	For cutting fabric, thread	Avoid handling from the sharp ends
Sewing machine	Used for stitching fabric pieces together	Operate only under supervision; keep fingers away from the needle
Pins	Used to hold fabric pieces in place before stitching	Insert and remove carefully
Iron	Used to press fabric and seams for neatness	Use under adult supervision; do not touch the hot surface
Needles	To stitch the cloth with thread	Avoid touching the pointed end
Thimble	Small, hard 'cap' worn on the finger to protect finger from needle pricks	Wear it on your finger used to push the needle into the cloth
Items for decoration	Such as lace, sequins	Could have sharp edges
Fasteners	Such as buttons, zips	Could have sharp edges
Tailor's chalk/Marker	To mark measurements and outlines	Use lightly so that it can be washed away

7.7 Bill of Materials

Fabric is generally available in the form of rolls. The width of a roll is typically between 1–1.5 m. The length of the cloth you need is based on the width of the roll and material you require.



Estimation





Cost estimation and documentation

Remember, only the measurements in your drawing are not sufficient for estimating the amount of cloth required. You need to estimate 'seam clearance', that is extra cloth, for stitching and hemming edges.

You should include the cost of labour and the estimated human hours spent doing the tasks. Such 'soft' aspects will help you understand the value of labour put in to complete the work.



CASELET

The students of Government High School created the following Bill of Materials for a child's dress (Table 7.6).

Table 7.6: Bill of Materials for apparel

Items	Quantity	Estimated cost (in ₹)
Main fabric (cotton)	1.5 m	300
Thread roll	1 roll	30
Button (for decoration)	10	20
Lace (for decoration)	0.5 m	10

Cost of labour	Value (time in hours × hourly estimate × frequency per week)	Estimated cost (in ₹)	Remarks (if any)
Measuring and sketching	1 × ₹50×1	50	
Stitching	2 × ₹50×2	200	
Total		610	



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Prepare the Bill of Materials for the apparel you will make.

7.8 Basics of apparel making

7.8.1 Sewing

The key part of sewing is handling the needle and thread. While you can stitch an apparel entirely by hand, some stitching by hand is required even when using a sewing machine. Thus, you should be able to use a needle and thread to carry out hand stitches competently.

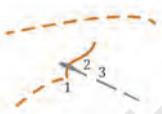
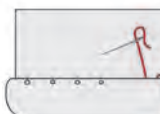
While hand stitching, you need to keep the following in mind (Figure 7.7 and Table 7.7).

1. Draw a reference line.
2. Decide the style of stitch in consultation with your teacher or the expert. Table 7.7 has a few examples of stitches and when to use them.
3. Keep the stitch straight and even.
4. Ensure consistent tension across all the stitches (neither too tight nor too loose).
5. Ensure no missed or uneven stitches.
6. Ensure no damage to the material being sewn.
7. Pull the needle away from yourself.



Figure 7.7: *Stitching by hand requires care*

Table 7.7: Types of stitches and when to use them

Stitches	Running stitch	Backstitch	Split stitch	Buttonhole stitch	Hemming stitch
Illustration					
When to use	Temporary stitching, joining layers, simple seams, e.g., stitching on an apron pocket.	Strong seams, repairs, outlines, e.g., stitching apron straps, repairing torn seams in a bag.	Embroidery, outlining designs, e.g., embroidery cushion cover designs or a dress.	For buttonholes, preventing fabric edges from fraying, e.g., buttonholes on a dress, securing edges of a bag flap.	Finishing hems of garments and fabrics, e.g., bottom edge of an apron, skirt hem.





Practise with a small sample



TASK

Practise different types of stitches on a sample cloth as shown in Figure 7.8. You can ask

experts to teach you stitches other than those in Table 7.7 or search for videos on the Internet. You can also try to repair any apparel or stitch small covers for water glasses from recycled cloth for practice.



Figure 7.8: Sampler with different kinds of stitches



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Place the sampler or an image of the sampler in your portfolio.



SAFETY

Basic safety while making an apparel using a sewing machine (Table 7.8).

Table 7.8: Safety while sewing



Place the machine on a flat, stable surface.



Excess portion of the fabric should hang on the left side.



Sit upright so that the centre of your body is aligned with the needle.



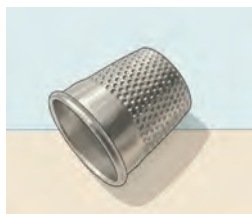
Do not pull the fabric while sewing, it automatically moves forward.



Light should come from the left side.



Cut the thread using scissors.



Use a thimble to protect your finger when pushing or pulling the needle through the fabric.



Use both hands while lifting or shifting the machine.

7.8.2 Drafting

In apparel making, the process of marking measurements and drawing the shape on paper is called drafting. Drafting helps in ensuring the right shape and size of the product, avoiding mistakes and reducing fabric wastage. Once the draft has been cut out, it becomes a paper pattern since it can be reused. The pattern is placed on fabric and traced to get the same shape and size on the fabric each time. Using a paper pattern helps in making the product accurate and neat, and getting the same measurements each time.

To develop a paper pattern for your product, follow the steps in Table 7.9.



Drafting and pattern making

Table 7.9: Developing paper patterns of apparel

Steps	What is to be done?
1.	Finalise the product design and note down all required measurements. These measurements are the actual measurements of the product, not reduced or enlarged as per any drawing scale. Unlike the technical drawing of construction, apparel patterns are drawn full size.
2.	Use chart paper or old newspaper, and draw the shape of the product using a scale and pencil, based on the actual measurements. Make sure to draw as per the design (e.g., square or round for the neck; rounded cutting for an armhole).
3.	Mark all measurements clearly on the paper pattern and label each part (front, back, strap, pocket, etc.).
4.	Recheck the paper pattern to ensure the size and shape are correct.
5.	Cut neatly along the drawn lines to prepare the paper pattern.





CASELET

Students of Grade 9 in Government High School started the process of drafting, making the paper patterns as shown in Figure 7.9.

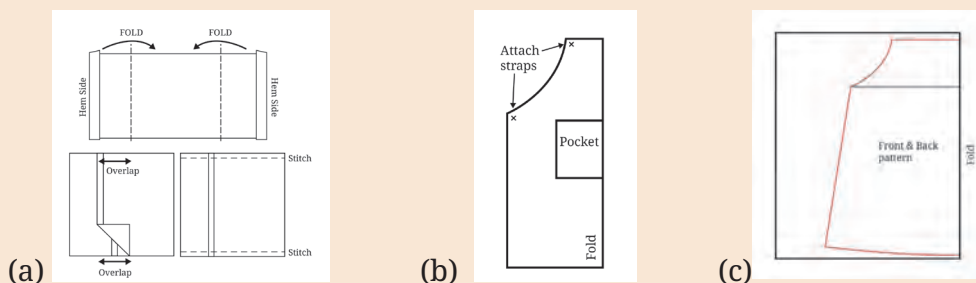


Figure 7.9: Paper patterns of different objects –
(a) cushion cover, (b) apron and (c) dress for a young child



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Prepare a paper pattern of the apparel you will make using chart paper or old newspaper.



Use of tools
and materials

7.8.3 Preparing fabric

After the paper pattern is ready, the next step is to prepare the fabric before cutting (Table 7.10). This step is crucial and should not be skipped.

Table 7.10: Preparing the fabric for stitching

Steps	What is to be included?	Image
1.	Pre-wash (and dry) your fabric before cutting (especially natural fibres like cotton, linen and rayon). Pre-washing fabric ensures that any shrinkage happens before you cut and sew.	
2.	Press the fabric using an iron to remove wrinkles and creases. Smooth fabric helps with accurate marking and cutting.	
3.	Fold the fabric so that the two side edges (called selvage) meet.	

7.9 Marking and cutting fabric

Once the pattern is ready, it can be used for cutting the fabric. Table 7.11 shows the key steps to be followed in this process.

Table 7.11: Steps for marking and cutting

Steps	What is to be done?	Image
1.	Marking of fabric as per the paper pattern: - Place the paper pattern straight along the length of the fabric; this ensures strength and durability of the apparel. Also, place the pattern so that there is minimum wastage of fabric (e.g., along the edges). - Pin the paper pattern on the fabric and trace around it with tailor's chalk, leaving 1–2 cm allowance for seams or hemming. - Take feedback from the teacher/expert before cutting.	
2.	Cutting of fabric as per the paper pattern: - Cut carefully along the marked lines on a flat surface. - Keep front and back pieces pinned together, and label them to avoid mix-up.	

QUALITY

Tips for cutting the fabric

- 1. Large, long strokes:** Make long, smooth cuts rather than many small snips.
- 2. Focus on the tip:** Look at the tip of the scissors to guide the direction of cutting for better accuracy.
- 3. Maintain blade parallel:** Keep the scissors parallel to the surface on which the fabric is placed.



Quality



PORTFOLIO

Place images or sketches of each of the steps you took to prepare, mark and cut the fabric in your portfolio. Write a brief description of the step.



CHECK YOUR UNDERSTANDING

Look at Figure 7.10 and answer the questions:

1. On the basis of what you learnt about marking and cutting of fabric, do you observe any errors in the figure?
2. What problems may occur due to these errors?
3. What steps should be followed to correct these mistakes?
4. State any two good practices for marking and cutting fabric. How will you apply these good practices in your work?



Figure 7.10: Uneven marking and cutting of fabric

Hint: Look at the placement of the paper pattern and the way the fabric has been cut.



Use of tools and materials

7.10 Sewing the product

After the fabric pieces have been cut and arranged correctly, the next stage is sewing them together. It is important to secure stitching at both the beginning and the end of the seam – this means that there must be no chance of the stitching coming loose at these points. This is done by making a few backstitches (that is, stitch ‘backwards’ over the initial few stitches), which help prevent the seam from opening during use or washing. Properly secured stitches increase the durability and life of the apparel.



SAFETY

While sewing, care should be taken to keep the stitches straight, even and properly aligned along the marked lines (Figure 7.11). Uneven stitching can affect both the appearance and strength of the product. Sewing should be done at a steady pace, so that the fabric does not slip (which will cause a few stitches to be missed) or gather (which will cause the fabric to become ‘scrunched’ up in some places).



Figure 7.11: Handstitching an apparel

CHECK YOUR UNDERSTANDING

In Figure 7.12, a few images are marked as incorrect while some are marked as correct. Do you agree? Give reasons for your answer.

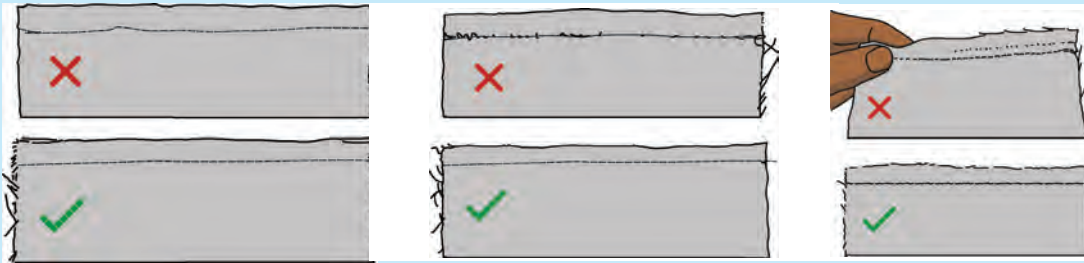


Figure 7.12: Samples of sewing

7.11 Finishing the product

Once stitching is complete, finish the raw edges of the seam allowance inside your garment by trimming or overlocking. This is done to prevent fraying, increase strength and ensure comfort. You will also need to neatly hem the bottom, sleeves and neck of the apparel. This is important for durability and a neat appearance. Unfinished edges, especially on woven fabrics, tend to fray with washing and wear. Seam finishing prevents this. Figure 7.13 shows simple methods to finish the raw edges of the seam allowance.



Finishing

Close hemming	Vertical hemming	Hand rolled hemming

Figure 7.13: Different methods of hemming



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Place a sketch or image of each step of the process in your portfolio. Briefly describe each step.

Did you need to redo the stitching? If yes, why and what did you do differently the second time?



DID YOU KNOW?

Surface Decoration

After stitching and finishing, you can use surface decoration to make your product more attractive or unique. Examples of some surface decoration techniques are embroidery, painting, patchwork, tie-and-dye, block printing, quilting and bead and lace or sequin work (Figure 7.14).



Figure 7.14:
Embroidery used for surface decoration

7.11.1 Packaging and labelling

After finishing and checking the apparel, the last step is to pack it neatly for delivery, storage or display. Packing is not just about wrapping the apparel. It involves careful thought about protecting it from dust, dirt and damage, and presenting it in a professional manner to the customer (Figure 7.15).



Figure 7.15: *Packing apparel for protection and presentation*

Similarly, labelling is important not only to display your brand name but also to inform the customer of the size of the apparel and how to ensure that it continues to look fresh. It is the primary means of communication between the manufacturer of the apparel and the customer. The label also helps the person selling the apparel to guide the customer. Figure 7.16 (a) provides some frequently used symbols, while Figure 7.16 (b) shows an example of a label.

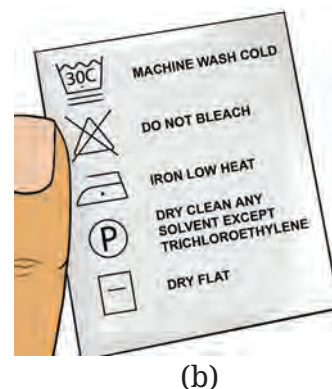
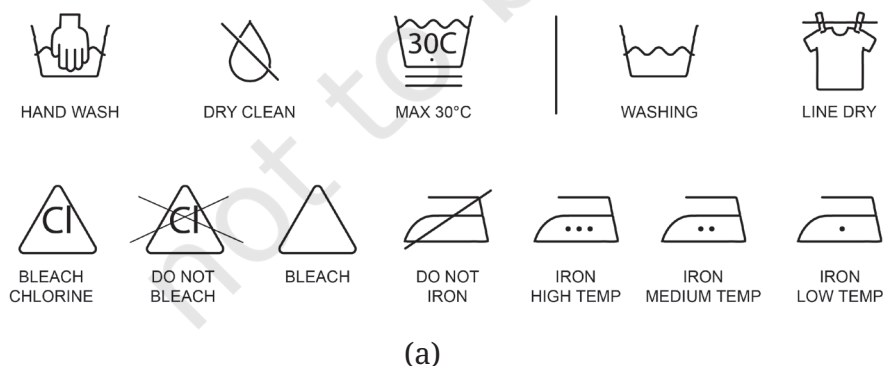


Figure 7.16: (a) *Frequently used symbols in labels of apparel and*
(b) *example of a label providing care instructions – the label cautions that trichloroethylene (a spot remover) cannot be used since it has carcinogenic properties.*



PORTFOLIO

Design a label for your apparel, sharing the size and care instructions. Attach it to the apparel and place the finished product or its image in your portfolio.

7.12 While making apparel

1. Before beginning any work, it is necessary to gather information related to the work. This can be done by visiting a tailoring shop or boutique, doing an online search or by interviewing a fashion designer or tailor.
2. Pre-wash, dry and press fabric before cutting to prevent shrinkage and uneven stitching.
3. Mark and cut fabric neatly and carefully, keeping pieces labelled to avoid confusion. Make long, smooth cuts rather than many small snips. Keep the scissors parallel to the cutting surface.
4. While using a sewing machine, sewing should be done at a steady pace so that the fabric does not slip or gather at once place.
5. Finish the product with hemming, trimming, pressing and checking quality.
6. Try to reuse materials, upcycle old clothes to reduce waste and protect the environment.

7.13 Assess your learning

1. Your task is to stitch a *pajama* set suitable for daily use. You can use cotton, polyester or a cotton-polyester, blend. Which fabric will you choose and why?
2. A learner takes measurements for (i) an apron and (ii) a dress while a person is wearing thick winter clothes. Explain how this may affect the final apparel and suggest steps for correction.
3. How, in your experience of making an apparel, does preparing a paper pattern before cutting help reduce fabric wastage?
4. Why is pre-washing and pressing fabric before cutting considered an essential step? What could happen if this step is skipped?



5. During stitching, a student notices uneven stitch length and that the stitching is loose. What problems could this cause during use and washing, and how can these issues be corrected?
6. Write down ten things you can do with your old dress. Find out what is sustainable fashion?
7. Of the tasks that you did, which did you enjoy the most? Which did you enjoy the least? Give examples of what went well and what did not go well. What would you do differently next time?
8. Give examples of how you can apply your learnings in a real-life situation.

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